Industrial Internship Report on

”Agriculture Crop Production Prediction in India using Machine Learning”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time. My project was **“Agriculture Crop Production Prediction in India using Machine Learning”**. The objective of the project was to build a machine learning model that can predict crop production based on multiple factors such as crop name, state, season, area, and cost. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

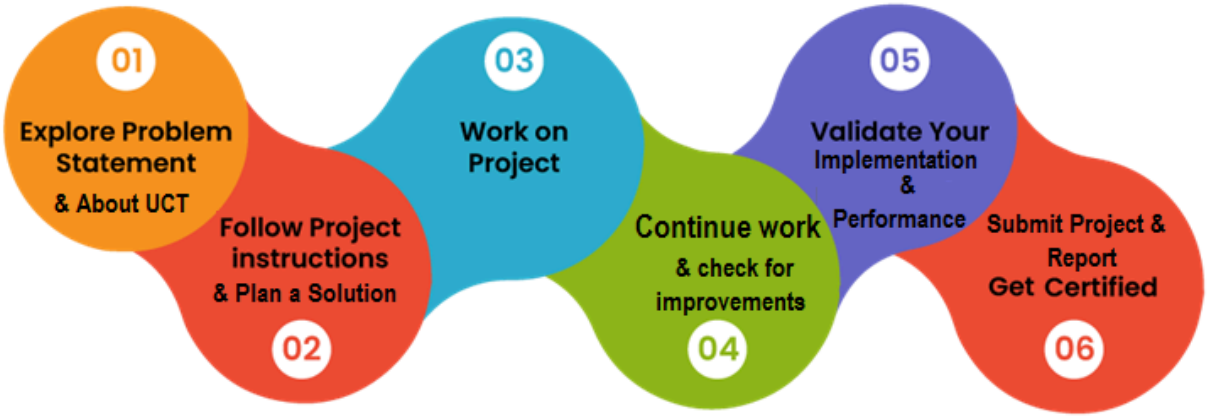
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

Thank to all (with names), who have helped you directly or indirectly.

Your message to your juniors and peers.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

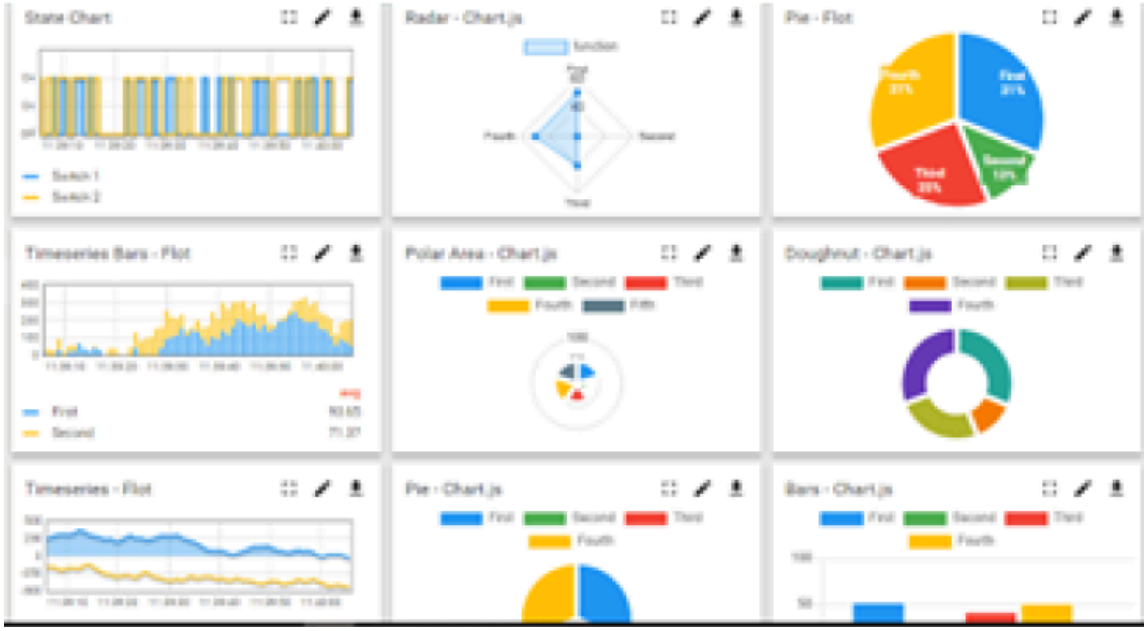
i. UCT IoT Platform (uct Insight)

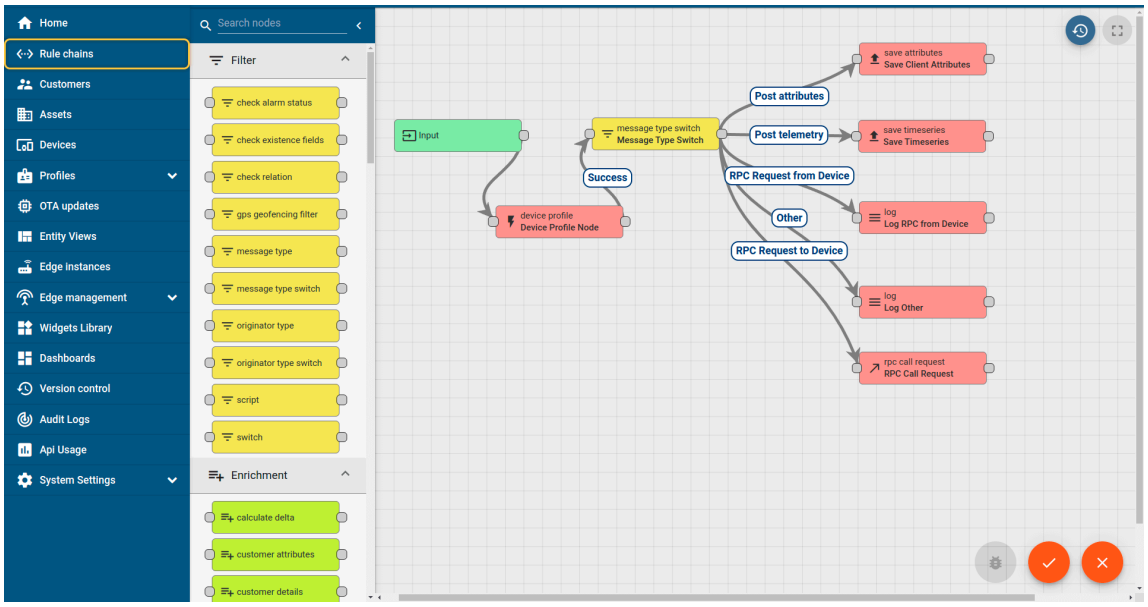
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA

- It supports both cloud and on-premises deployments.

It has features to
• Build Your own dashboard
• Analytics and Reporting
• Alert and Notification
• Integration with third party application(Power BI, SAP, ERP)
• Rule Engine


State Chart
Radar - Chart.js
Pie - Flot
Timeseries Bars - Flot
Polar Area - Chart.js
Doughnut - Chart.js
Timeseries - Flot
Pie - Chart.js
Bars - Chart.js


Home
Rule chains
Customers
Assets
Devices
Profiles
OTA updates
Entity Views
Edge instances
Edge management
Widgets Library
Dashboards
Version control
Audit Logs
Api Usage
System Settings
Search nodes
Filter
check alarm status
check existence fields
check relation
gps geofencing filter
message type
message type switch
originator type
originator type switch
script
switch
Enrichment
calculate delta
customer attributes
customer details
Input
device profile
Device Profile Node
Success
message type switch
Message Type Switch
Post attributes
Post telemetry
RPC Request from Device
Other
RPC Request to Device
save attributes
Save Client Attributes
save timeseries
Save Timeseries
log
Log RPC from Device
log
Log Other
rpc call request
RPC Call Request

ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

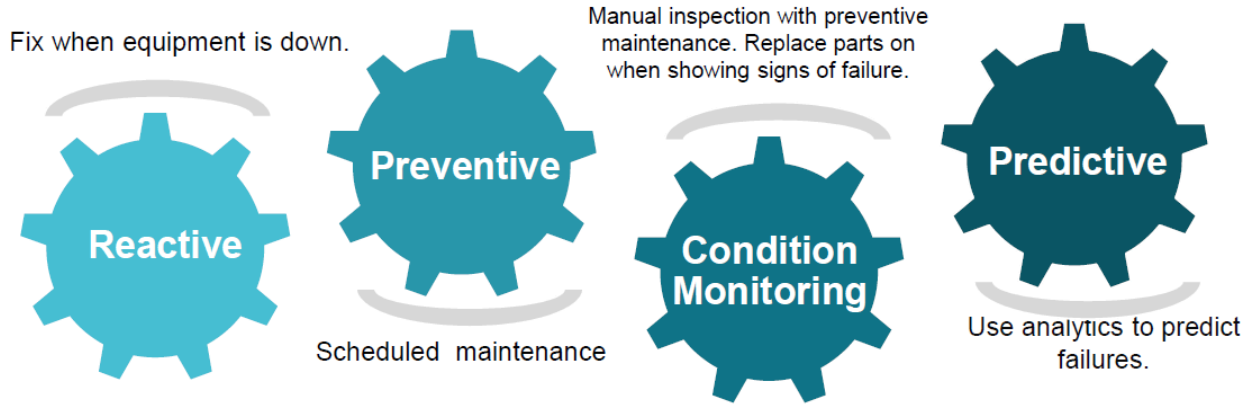
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

upSkill CAMPUS

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upSkill CAMPUS

Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com

Professional networking
- Interview Preparation
- Alumni Connections
- Upskilling Courses
- Mentorship
- Skill Assessment
- Discussion/QA forum
- Profile building

Collaboration platform
- Project collaboration
- Discussion forum
- Tech updates

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Job/internship platform
- Job portal
- Internship portal
- Freelancing projects

2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] upSkill Campus Internship Program Guidelines

[2] UCT Official Website and Project Problem Statement

[3] Python Documentation

[4] Scikit-learn Documentation

2.6 Glossary

Terms	Acronym
ML	Machine Learning
EDA	Exploratory Data Analysis
CSV	Comma Separated Values
R2 Score	Coefficient of Determination
MSE	Mean Squared Error
Random Forest	Ensemble Learning Algorithm

3 Problem Statement

Agriculture plays a major role in the Indian economy and is a primary source of livelihood for a large population. Accurate crop production prediction is very important for farmers, government bodies, and agricultural industries for proper planning of storage, distribution, pricing, and export decisions.

However, crop production depends on many factors such as crop type, state, season, cultivated area, cost of cultivation, and other environmental conditions. Traditional methods of estimating crop yield are often based on experience or manual calculations, which may not provide accurate results.

Therefore, there is a need for an automated prediction system that can analyze past agricultural data and provide reliable production predictions.

The problem statement of this project is:

"To develop a machine learning-based system that predicts crop production in India using historical agricultural dataset features such as crop name, state, season, area, and cost."

The goal is to build a regression model that can learn patterns from historical data and predict future crop production values, which will help in improving agricultural planning and decision-making.

4 Existing and Proposed solution

#Summary of existing solutions provided by others, what are their limitations?

In many rural and semi-urban areas, farmers and agriculture organizations estimate crop production using traditional methods. These methods are mostly dependent on:

- manual calculations
- previous year production records
- local knowledge and farmer experience
- weather assumptions and seasonal estimations

In some cases, basic statistical approaches such as average production calculation are used. However, these approaches have several limitations:

- Predictions are not accurate due to lack of proper data analysis
- Manual prediction is time-consuming and requires human effort
- Traditional estimation cannot handle large datasets
- Seasonal and state-wise variations are difficult to calculate manually
- It does not provide fast decision-making support

Due to these limitations, there is a requirement for an intelligent system that can provide better predictions using data-driven techniques.

#What is your proposed solution?

The proposed solution is to develop a machine learning model that predicts crop production in India based on historical agricultural data.

In this project, the dataset includes features such as:

- Crop name
- State
- Season
- Area / Quantity
- Cost
- Production (Target Variable)

The system performs the following tasks:

- Loads the dataset in CSV format
- Performs data cleaning by removing missing values and duplicate records
- Performs Exploratory Data Analysis (EDA) using graphs to understand patterns
- Applies feature engineering techniques such as label encoding for categorical columns
- Splits the dataset into training and testing data
- Trains a regression model (Random Forest Regressor / Linear Regression)
- Evaluates model performance using metrics such as R2 Score and MSE
- Provides a Streamlit-based user interface for prediction

#What value addition are you planning?

The proposed system provides the following advantages:

- Automated crop production prediction
- Faster and more accurate results compared to manual estimation
- Helps farmers and organizations make better decisions
- Can be improved in future by adding weather and soil-related parameters

Thus, the proposed machine learning-based solution provides an efficient and scalable method for crop production prediction.

4.1 Code submission (Github link):https://github.com/rainar2005-dev/upskillcampus

4.2 Report Submission (GitHub link):

https://github.com/rainar2005-dev/upskillcampus/blob/main/Agriculture_Crop_Production_Prediction_Raina_USC_UCT.pdf

4.3 Comparison of Existing and Proposed Solution

The existing solution for crop production prediction mainly depends on manual estimation and traditional statistical methods. These approaches are time-consuming and may not provide accurate results because they cannot handle large datasets and complex patterns in agricultural data.

The proposed solution uses machine learning algorithms such as Random Forest Regression and Linear Regression to predict crop production. This approach is faster, more reliable, and provides better accuracy by learning patterns from historical data. The proposed system also supports automation and can be easily enhanced in the future by adding more features like rainfall, temperature, and soil type.

5 Proposed Design/ Model

The proposed system is designed to predict agriculture crop production in India using Machine Learning. The design flow starts with uploading a crop production dataset in CSV format. After uploading, the dataset is cleaned by removing missing values and duplicate records to improve data quality. Then Exploratory Data Analysis (EDA) is performed to understand production patterns across states, crops, and seasons using graphical visualizations.

After data analysis, feature engineering is applied where categorical values such as crop name, state, and season are converted into numerical values using encoding techniques. The cleaned dataset is then split into training and testing sets. A Machine Learning regression model such as Random Forest Regressor is trained on the training dataset and evaluated on the testing dataset using evaluation metrics like R2 Score and Mean Squared Error (MSE).

Finally, the trained model is integrated into a Streamlit web application where the user can input values such as crop name, state, season, and area to predict the expected crop production output. The final outcome of the system is a predicted crop production value along with graphical insights and model evaluation results.

5.1 High Level Diagram

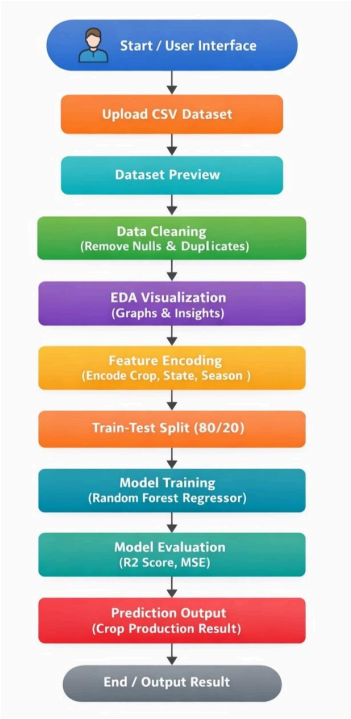

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

High Level Flow:

- Upload Dataset
- Data Cleaning
- EDA Visualization
- Feature Encoding
- Train-Test Split
- Model Training
- Model Evaluation
- Prediction Output

5.2 Low Level Diagram

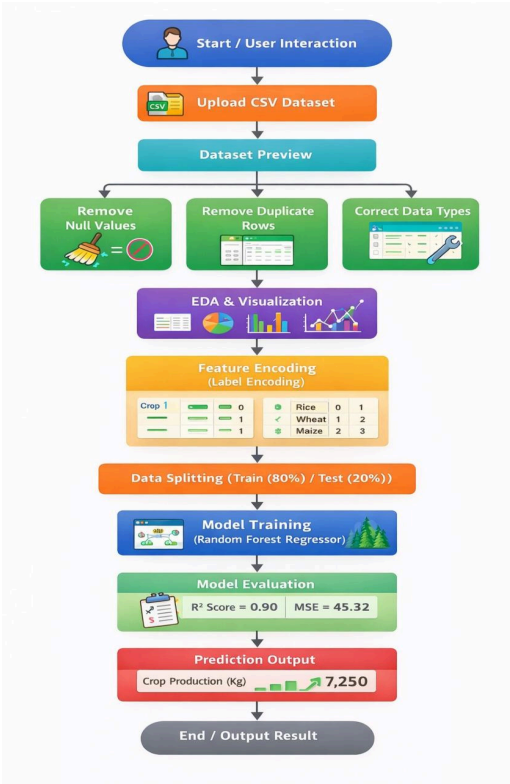

Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

Low Level Steps:

- Import libraries (pandas, numpy, sklearn, matplotlib)
- Load CSV dataset
- Handle missing values and duplicates
- Perform visualization for analysis
- Convert categorical values to numeric (Label Encoding)
- Split dataset into training and testing sets
- Train Random Forest Regressor model
- Evaluate using R2 Score and MSE
- Build prediction function for user input
- Display results on Streamlit UI

5.3 Interfaces (if applicable)

The project uses **Streamlit** as the user interface.

The Streamlit app includes:

- Dataset Upload option
- Dataset Preview
- Graphs for analysis
- Model training and evaluation output
- Prediction form

6 Performance Test

Constraints Identification (Industrial Perspective)

In real industries, machine learning models must work efficiently with large datasets and should provide accurate predictions within limited time and computing resources. While developing this crop production prediction system, the following major constraints were identified:

1. Dataset Quality Constraint

Agriculture datasets often contain missing values, duplicate records, and inconsistent entries due to manual data collection. Such data quality issues can reduce model performance and lead to incorrect predictions.

How it was handled:
This constraint was managed by applying data cleaning techniques such as removal of null values and duplicates.

Test Result:
After cleaning, the dataset became consistent and model training was performed successfully without errors.

2. Accuracy Constraint

In industrial agriculture forecasting, prediction accuracy is critical because wrong estimation may lead to financial loss, poor resource planning, and incorrect market decisions.

How it was handled:
A Random Forest Regressor model was selected because it provides better performance on complex datasets compared to simple regression models.

Test Result:
The trained model produced acceptable prediction accuracy measured using evaluation metrics such as R2 Score and Mean Squared Error.

3. Speed and Computation Constraint

Machine learning training can take significant time when the dataset is large, especially when multiple models are tested.

How it was handled:
The dataset was processed efficiently using Python libraries and the model was trained using optimized Scikit-learn implementation. Train-test split was applied to ensure quick evaluation.

Test Result:
The model was trained successfully within reasonable time and predictions were generated instantly through the Streamlit interface.

4. Scalability Constraint

In real-world applications, crop production datasets can increase in size over time and may include additional factors like rainfall, soil type, temperature, and fertilizer usage.

How it was handled:
The project design is modular (separate model.py and app.py), making it scalable and easy to update with new features and models.

Recommendation:
For large industrial datasets, cloud platforms like AWS/Azure can be used to improve computation speed and scalability.

5. Model Generalization Constraint

A machine learning model may perform well on training data but may fail when new unseen data is provided.

How it was handled:
Train-test splitting and evaluation metrics were used to ensure the model performs well on unseen test data.

Recommendation:
Future improvement can include cross-validation and hyperparameter tuning to enhance generalization.

Final Conclusion on Constraints

The identified constraints were addressed through data preprocessing, selection of suitable machine learning models, evaluation metrics, and modular project architecture. These steps ensure the project is suitable for real-world industrial applications and not limited to academic implementation.

6.1 Test Plan/ Test Cases

Test Case	Input	Expected Output
Upload CSV	Agriculture dataset	Datasets loads successfully
Cleaning	Dataset with null values	Null values removed
EDA Graph	Crop column	Graph displayed correctly
Model Training	Training datasets	Model trains without error
Evaluation	Test dataset	R2 score and MSE generated
Prediction	User inputs crop/state/season	Production prediction output

6.2 Test Procedure

- Ran the Streamlit application using streamlit run app.py.
- Uploaded the agriculture dataset (CSV file).
- Verified dataset preview output.
- Performed data cleaning (removed null values and duplicates).
- Generated EDA visualizations (crop-wise, state-wise, season-wise graphs).
- Applied label encoding to categorical columns (Crop, State, Season).
- Split dataset into training and testing sets (80/20).
- Trained the model using Random Forest Regressor.
- Evaluated the model using R2 Score and MSE.
- Tested prediction form with sample inputs and verified predicted output.

6.3 Performance Outcome

The model was evaluated using regression evaluation metrics:

- **R2 Score**: Indicates how well the model explains the variance in crop production
- **MSE (Mean Squared Error)**: Measures average squared difference between predicted and actual output

The results were displayed through:

- R2 Score value
- MSE value
- Actual vs Predicted graph

The model successfully predicted crop production values based on the input parameters.

7 My learnings

During this internship, I gained practical knowledge of implementing a complete machine learning project from scratch. The key learnings include:

- Learned how to work with real-world datasets and understand their structure.
- Improved skills in **data preprocessing** such as handling missing values and duplicate entries.
- Gained experience in performing **Exploratory Data Analysis (EDA)** using visualizations to identify trends and patterns.
- Understood the importance of **feature engineering** and encoding categorical variables for machine learning models.
- Learned how to split dataset into training and testing sets for model evaluation.
- Implemented machine learning regression models such as **Random Forest Regressor** and **Linear Regression**.
- Learned evaluation techniques using **R2 Score** and **Mean Squared Error (MSE)**.
- Gained hands-on experience in building a user-friendly web application using **Streamlit**.
- Understood the importance of GitHub for maintaining project code and documentation.
- Improved technical skills in Python libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn.

Overall, this internship enhanced my confidence in machine learning and helped me understand how industrial projects are structured and implemented.

8 Future work scope

The current project provides a basic crop production prediction system using machine learning. However, the system can be improved further by adding more features and enhancements:

- Include additional parameters such as **rainfall, temperature, humidity, soil type, and fertilizer usage** for better prediction accuracy.
- Apply advanced machine learning algorithms such as **XGBoost, Gradient Boosting, and Neural Networks**.
- Perform **hyperparameter tuning** to improve model performance.
- Add **cross-validation** for better generalization of the model.
- Create a crop recommendation system that suggests the best crop based on input conditions.
- Deploy the application on cloud platforms like **AWS or Azure** for real-time usage.
- Build an advanced dashboard with more interactive charts and filters.
- Add state-wise and season-wise prediction reports for better decision making.

These improvements will make the project more scalable and suitable for real-world agriculture industries.